TFE4152 Design of integrated circuits

"Pacemaker Project", fall 2026
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8th of October, 2026, v1 (Updated Oct. 8)

1 Some brief information about pacemakers, relevant for the project

Implants include electronic systems exploiting integrated circuit technologies that often have very stringent requirements with respect to low power / low energy implementation, like for example the wireless sensor node called the world's smallest computer [1], which measures maximum 0.3 mm along any among it's sides. It has been investigated as a tool for detecting cancer [1], measuring pressure inside the eye for glaucoma diagnosis, biochemical process monitoring, and a wide range of other applications [2]. Such tiny sensor nodes may use a wide variety of different technologies and computational substrates [1], [3]. Some more technical details about the circuitry from University of Michigan have been published in [4].
One type of implant that have a long history, is the pacemaker, which have evolved into ultra low power tiny devices, from origins including large bulky mechanical objects.
Today's pacemakers can be internet connected reprogrammable healt management systems. This project task is inspired by some basic functionalities of such devices, being very simplified though, to limit the scope of the project.

A brief description of common present day pacemakers can be as follows [5]:

A pacemaker is an implanted device that uses electrical pulses to keep the heart beating at a regular rate. They are used to treat heart failure, and abnormal heart rhythms which can cause patients to lose consciousness. The device, which consists of a small metal box containing a battery and a small computer, sits under the skin near the collarbone. Leads attached to the box are threaded through a blood vessel to the heart's chambers. The pacemaker senses via these leads what the heart is doing, and to add in beats if necessary. If the heart slows down or misses a beat, the pacemaker sends an electrical impulse, stimulating the heart to restore it to its normal rate.
There are several different types, which offer a range of sophisticated options. These include regulating a slow heart rate; prompting the heart chambers to beat in time; increasing the heart rate when the patient is active; remote monitoring, and activity pattern storage. Not all of these options are needed by every patient.

The heart could beat too fast (tachycardia), too slow (bradycardia), or in irregular ways. Irregular heart beats may be classified as arrhythmia. Two types of pacemakers are illustrated in Figure 1. One is a dual-chamber pacemaker, that includes an electronic device connected through leads / wires to the right atrium and the right ventricle, respectively. This helps to keep the pace of contractions of both chambers, when needed [6]. The other type is a leadless pacemaker, not needing wires, connected to the inner heart wall. Electric signals produced by the heart are measured and amplified, to enable decisions regarding whether the heart should be paced or not. A healthy heart produces results from such electric signals, the QRS complex, depicted in Figure 2.

For people of the age of 75 and above, about 1 in 50 persons has a pacemaker implant. Pacemakers [7] may increase quality of life, prevent or avoid unpleasant symptoms, or ultimately save lives.
This project evolved from the idea of some functions relevant for taking care of a heart beating too slow (bradycardia), and a single-chamber variant, which will be further described.

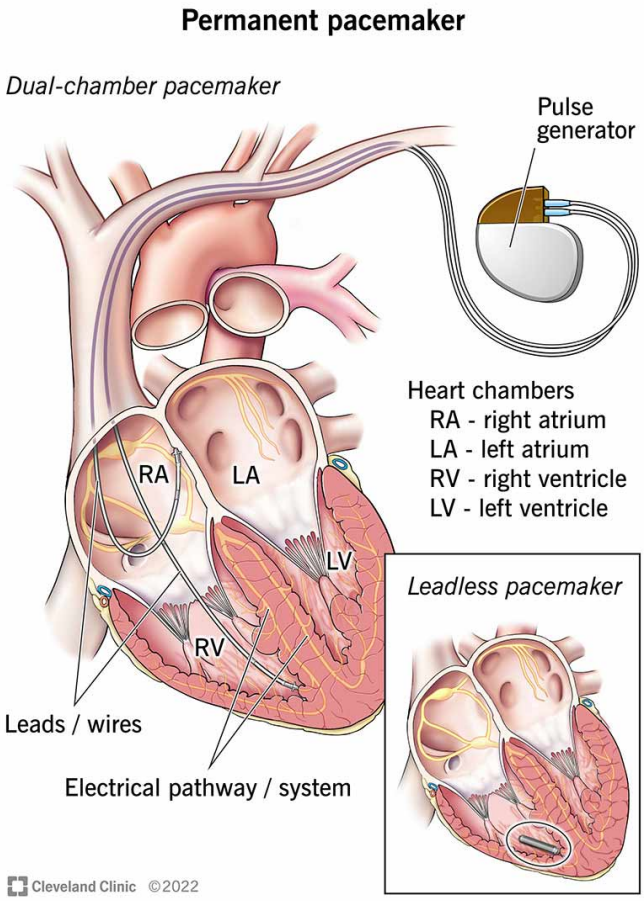


Figure 1: Dual chamber / Dual lead pacemaker, electrode, and Single-chamber / Single lead pacemaker [8]

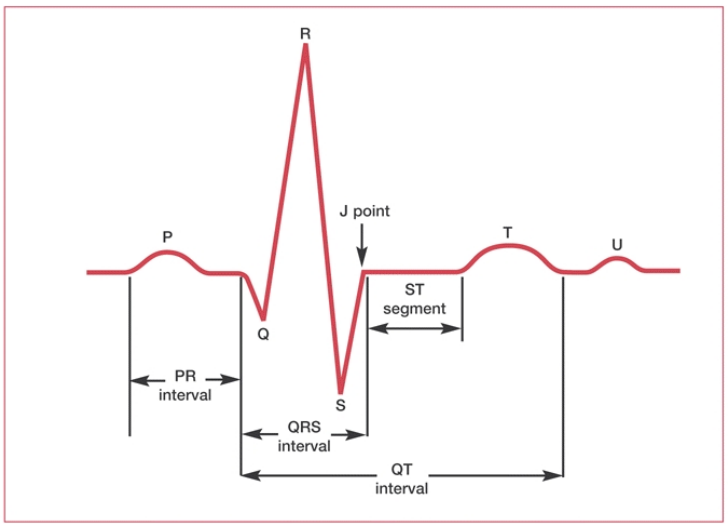


Figure 2: QRS complex [7]

2 Functionality relevant for the single chamber pacemaker

2.1 Brief, overall description of the single chamber relevant implementation

The first part of the project is inspired from the design of functionality relevant for a single-chamber pacemaker, illustrated in Figure 3 c). A significant fraction of people having a pacemaker, have a variant of this type. We imagine that it uses an electrode for sensing heart activity as well as pacing the heart. The measured signal from the heart is processed by a **sensing amplifier**, and any heartbeat detected by a digital control block.
In commercially available pacemakers the signal processing chain is typically involving more advanced circuitry in the analog signal processing path, like for example V to I converters, bandpass filters, adaptive threshold circuits, comparators and analog-to-digital converters [9].
The **digital control block** decides when the heart is going to be paced. In the cases where the heart is paced, an **output driver** provides the electrical signal to the electrode that is connected to the heart chamber.
The pacing should in this case only happen if the heart isn't beating within a certain time interval since the previous such event. Consider 60 beats per minute a normal resting heart rate, even though this varies from one individual to the next, as well as over time, for one and the same individual.
You may assume that the digital control, which should be implemented as a Moore-type Finite State Machine, includes a state where it waits for the sensing amplifier to sense a heart beat, resulting in a Boolean input to this digital control. (A general FSM is depicted in Figure 4.) This should happen if the heart rate drops below about 40 beats per minute. If there is a need to pace the heart, the digital control should change an output to a logic 1 for a short time, which would activate circuitry stimulating the heart to beat, via the electrode -the output driver.
For your state machine, you shall have boolean input for registering that a heartbeat has been sensed, "H", and at least an output signal indicating if the heart needs to be paced,"P", which in such case should be set to a logic 1 for a relatively short time.
In addition to this, you can use a square wave signal having a frequency of 512 Hz to extract a clock signal for the FSM, for example.
Figure 6 shows the main digital part, including a timer, receiving a reset signal, R, and providing a high, T, signal when the time has run out, and the heart should be considered to pace the heart. You may choose to use the 512 Hz signal as input to the timer, if you decide to.
Further requirements will be provided later in this document.

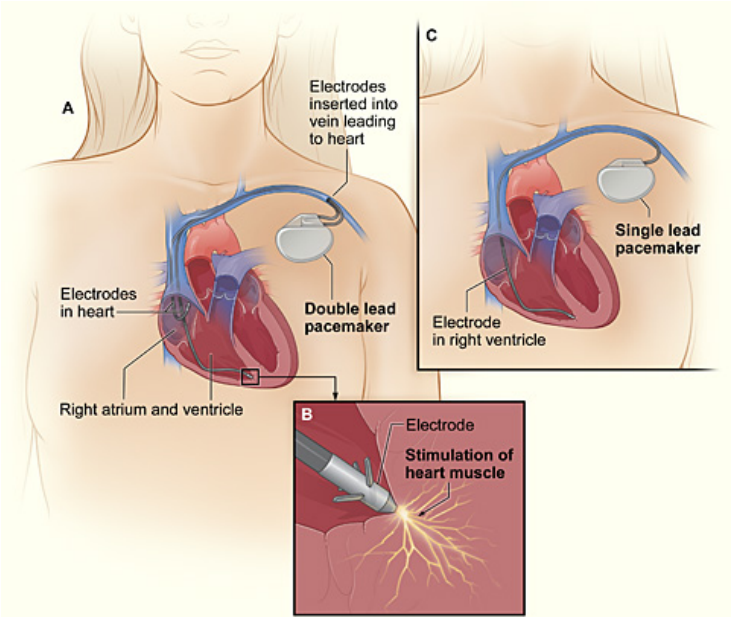


Figure 3: Dual chamber / Dual lead pacemaker, electrode, and Single-chamber / Single lead pacemaker [7]

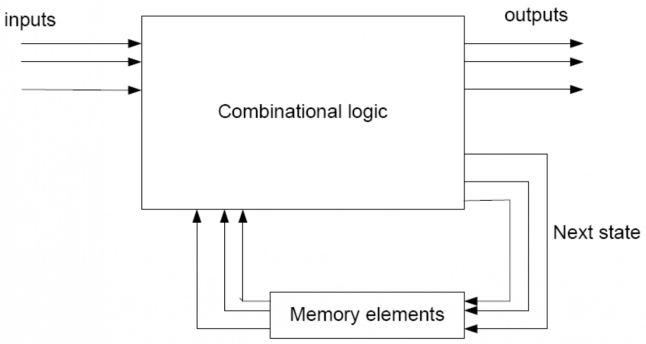


Figure 4: Finite State Machine

2.2 Requirements for the digital parts

You should model the behaviour of the digital part mentioned above in Verilog. A Moore machine model must be used for the implementation of a Finite State Machine (FSM) describing it's behaviour. A top-level view of a Finite State Machine is shown in Figure 4.
The beginning of a state table to be used for your design is shown in Table 1. In addition to the outputs,defined in the table (P,R), you may add additional outputs that you think may make sense.
The code should be written at a gate level, where every logic gate and interconnects are traceable down to the transistor level. By traceable down to the transistor level it's meant that it's explicit how the logic building blocks like NAND, INV et cetera used for the gate level code would have their inputs and outputs connected to the abstraction level below gate-level verilog, namely transistor schematics for the implementation of a 2-input NAND, for example. (Whether a certain input signal is provided to the upper or lower among two NMOS transistor in a NAND2, for example, would mean different electrical properties of the total circuit that could sometimes be relevant.)
The gate-level code requirement does not apply to any Verilog for test benches. (The course is focusing on low power implementations, and the use of higher abstraction levels could detach your design from possibilities to reflect upon, and implement, effective low power / low energy solutions.) It might be useful to look at the pen-and-paper method dealt with in lectures, as an inspiration for your own approach.
Verilog using Active HDL must be used.

Current State	Current State	input	Input	Next State	Next State	Output	Output	Output x,y,z?
MSB	LSB	H	T	MSB	LSB	P	R	
0	0	0	0					
0	0	0	1					
0	0	1	0					
0	0	1	1					
0	1	0	0					
0	1	0	1					
0	1	1	0					
0	1	1	1					
1	0	0	0					
1	0	0	1					
1	0	1	0					
1	0	1	1					
1	1	0	0					
1	1	0	1					
1	1	1	0					
1	1	1	1					

Table 1: State Table



Figure 5: View of the system on a high abstraction level.

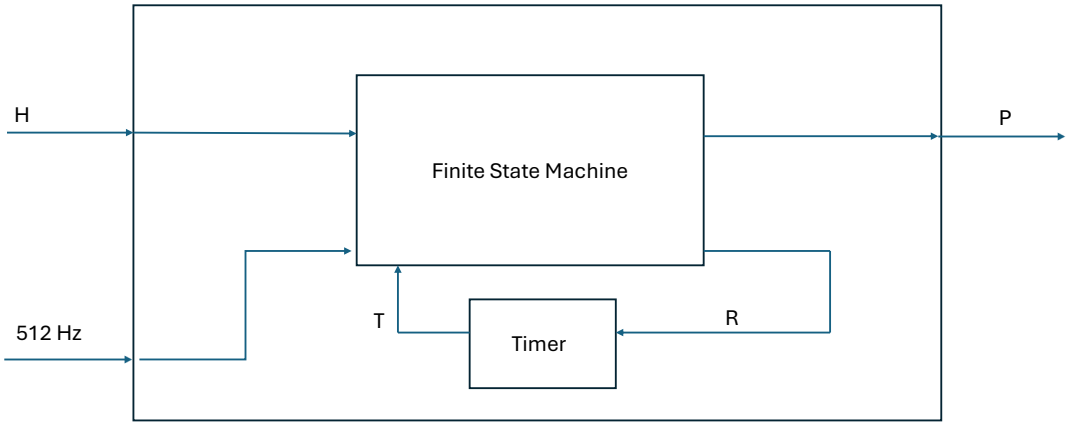


Figure 6: Block diagram for the digital part of the project.

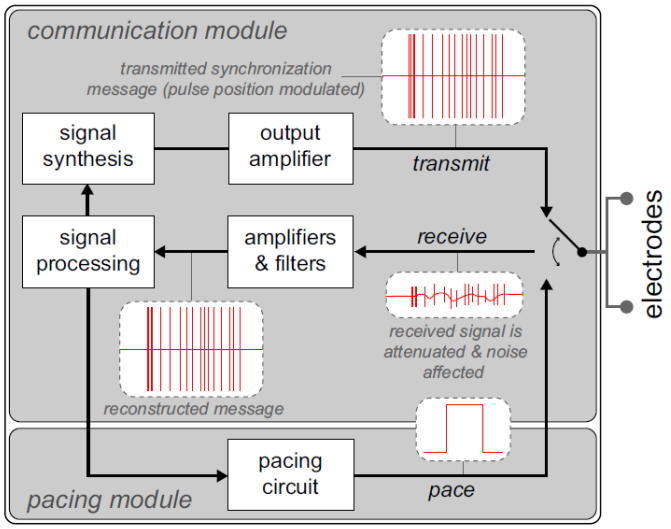


Figure 7: Leadless pacemaker described in [10].

2.3 Requirements for the analog parts

For any Spice simulations, AIMSpice should be used. The process technology is the 0.18μm CMOS used in "CJM" [11].

2.3.1 Analog circuitry involved in the front-end

Analog circuits for pacemakers can take different forms, while a diagram for a leadless pacemaker that can be connected to one or more other such, is shown in Figure 11.

Assume that the QRS complex (Figure 2) that has a max R value of 5 mV has been amplified, filtered, and level shifted, so that the ST segment lies at a DC-level of 70 mV, while the maximum R value reaches an additional 90 mV (total of 160 mV referred to ground). The P, Q, T and U components have been filtered out, and are replaced by a similar DC-level as that of the ST segment. Ensure that the relevant parts of the QRS complex are recognized as heart beats ("H") by your Finite State Machine, by amplifying the voltage swing of the previously described QRS complex. The digital FSM which is implemented in 180 nm CMOS should use a single supply voltage upwards limited to 1.8 V.
You are allowed to use maximum 1 independent source (voltage or current) to support the analog front-end functionality.
Any CMOS building blocks should be implemented using AIMSpice and as default build on the TFE4152 curriculum.

2.3.2 Analog circuitry involved in the pacing of the heart

Design an analog circuit being able to pace the heart with a current for approximately 2 ms +/- 50% producing an maximum amplitude of 1.5 V (recommended: 1.0 V-1.5 V) through a 500 Ohm resistor to ground (0 V), mimicking the typical impedance that the pacing pulse opposes.
You are allowed to use maximum 1 fixed independent source (voltage or current) to support the analog pacing functionality.

2.3.3 Additional requirements for the full project

To limit the scope, you should focus on implementing the combinatorial part of your state machine for the least possible static power consumption. Then it's time to step down in abstraction level and simulate this combinatorial part using AIMSpice and transistor level netlists. Normally one should take Process-, Voltage- and Temperature variations into account when performing analog / AIMSpice simulations. For simplicity we restrict the requirements to process variations only. As a rule of thumb, adequate logic 0 and 1 signals may have the following "worst case limits": for logic 1; down to 0.75 V_{dd}, for logic 0; up to 0.25 V_{dd}.
The IMRaD format is often used for scientfic works, and useful principles that may be useful also for you, may be found in descriptions like [12] and https://guides.lib.uci.edu/scientificwriting/home .
The full project report should be finished by the *Friday 20th of November, 14:00*.

Updates to this document may occur, but this version is distributed to give you a possibility to consider what it's about, at this point in time (2026.10.08).
Additional guidelines and requirements for the report, apart from those described here, will be provided.

2.4 Acknowledgements

Thanks to Erik Lewerenz and Jonathan Sæthre Ege for fruitful discussions and input related to the creation of the project description.

References

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3 Appendix - Pacemaker variants

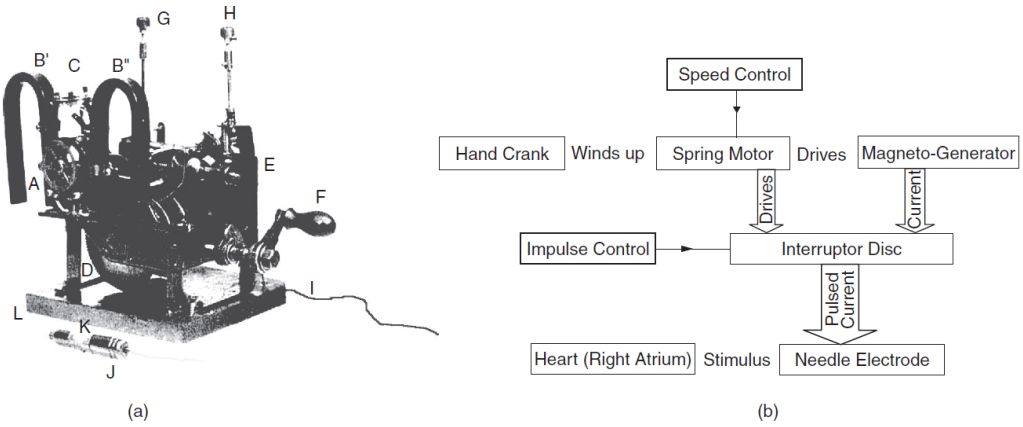


Figure 8: Hyman's pacemaker [15]

The first pacemaker was made by Australian anaestheseologist Albert Hyman, and is depicted and illustrated in Figures 8 a) and b). P. M. Zoll pioneered the electrical pacemaker, which weighted several kilos, tended to burn the skin, and was connected to 110 V AC, and may be seen in Figure 9 [13].
Single-chamber pacemakers use a single wire to one of the heart chambers [8], while Dual-chamber pacemakers use two wires to two different heart chambers [8]. Illustrations may be found in Figure 1.
Biventricular pacemakers use 3 wires, where 2 are connected to the lower heart chambers (ventricles) and the third to an upper chamber (right atrium), for the purpose of cardiac resynchronization therapy (CRT) [8].
Traditional electronic pacemakers include electrodes that sense the natural heartbeat, receiving signals that are amplified. A pulse generator creates the electrical pulses that are delivering pulses to the heart, when needed, through wires (also called leads).



Figure 9: Zoll's E-11 pacemaker [13].

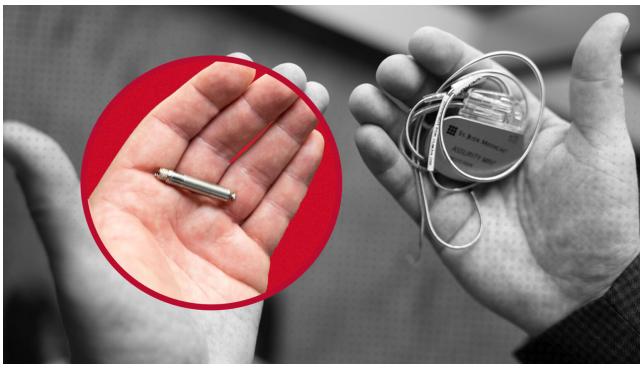


Figure 10: Leadless pacemaker to the left, and a traditional one to the right [**?**].

Present day pacemakers are often reprogrammable, and include circuitry for digital control, processing and memory, as well as wirless communication. Multi-modal sensing capabilities to make them adapt their functionality to the requirements and changing conditions of the user, are also included in many. The user may have data from the pacemaker read out to be scrutinized so that parameters for the pacemakers might be updated depending on needs. A couple of modern variants are shown in Figure 10.



Figure 11: *"Last Monday, I had surgery to become a little bit more of a machine: I got a pacemaker,"* Schwarzenegger stated [14].